SOFTWARE REQUIREMENTS SPECIFICATION

STUDENT RECORD SYSTEM

Version 1.0 | Programming Language: C / C++

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Revision History

Version	Date	Author(s)	Change Summary
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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional, non-functional, and security requirements of the Student Record System (SRS). The system provides a centralized, reliable environment for maintaining student academic records, including grades, attendance, personal details, and performance reports, for use by academic institutions.

1.2 Scope

The platform covers student registration and profile management, grade entry and computation, attendance tracking, report generation, and role-based access for students, faculty, and administrators. It is implemented in C/C++ and does not cover financial transactions, online learning modules, or live examination proctoring.

1.3 Audience

Students, faculty members, administrators, developers, testers, and project evaluators.

1.4 Definitions

Term	Definition
Student Record	A persistent data entry containing a student's personal, academic, and attendance details.
Grade	A numeric or letter value assigned to a student for a course based on assessment performance.
Attendance	A daily or per-session record of a student's presence or absence in a course.
Faculty	An authorized instructor who can enter and manage grades and attendance for their courses.
Administrator	A privileged user responsible for managing user accounts, courses, and system configuration.
GPA / CGPA	Grade Point Average / Cumulative GPA, computed from course grades and credit weights.
Report	A structured summary of a student's academic and attendance data for a given period.

2. Overall Description

2.1 Product Perspective

The Student Record System is a standalone C/C++ application with a command-line or lightweight GUI interface and a file-based or embedded relational database for persistent storage. It replaces manual registers and scattered spreadsheets with a unified system in which student profiles, grades, attendance, and reports are consistently managed.

2.2 Major Product Functions

- Student registration, profile creation, and personal data management
- Course and section management
- Grade entry, modification, and GPA/CGPA computation
- Attendance recording and percentage calculation
- Academic report generation (per student, per course, per semester)
- Role-based access control for students, faculty, and administrators
- Search and filter across student records

2.3 User Roles and Characteristics

Role	Main Capabilities
Student	View own profile, grades, attendance records, and generated reports.
Faculty	Enter and update grades and attendance for assigned courses; view course-level summaries.
Administrator	Manage all student profiles, user accounts, course assignments, and system data; generate institution-wide reports.

2.4 Operating Environment

- Desktop computer running Windows 10/11 or Linux (Ubuntu 20.04+)
- C/C++ compiled application (GCC / MSVC toolchain)
- File-based storage (CSV/binary) or SQLite for persistent data

- Minimum: 2 GB RAM, 100 MB disk space

2.5 Constraints

- Only authenticated and authorized users shall perform protected operations.
- All student data must be validated before storage (format, range, required fields).
- GPA/CGPA computation depends on correctly configured course credit weights.
- The system requires access to the data storage layer for normal operation.
- Passwords must be stored as hashes; plaintext storage is prohibited.

3. External Interface Requirements

3.1 User Interfaces

- Login screen (username / password)
- Student dashboard — profile, grades, attendance summary
- Faculty dashboard — course list, grade entry, attendance marking
- Administrator dashboard — user management, course management, reports
- Search and filter panel for student records
- Report viewer / export screen

3.2 Hardware Interfaces

No specialized hardware is required. The system shall operate on a standard desktop or laptop computer.

3.3 Software Interfaces

The C/C++ application shall interface with a file system or SQLite database for persistent storage. Export functions shall produce CSV or plain-text report files readable by standard office tools.

3.4 Communication Interfaces

The initial version is a local desktop application with no network communication. If a networked version is required in a future phase, standard TCP/IP sockets and TLS shall be used.

4. System Features

Each functional requirement includes acceptance criteria and a test-case reference. Requirement IDs follow the format SRS-F-###.

4.1 Authentication & User Management

Req ID	Requirement	Priority	Acceptance Criteria / Test Case	Dependencies
SRS-F-001	The system shall allow a registered user to log in using a valid username and password.	High	AC: Valid credentials grant access; invalid credentials display an error. Test: TC-Auth-01	All roles

Req ID	Requirement	Priority	Acceptance Criteria / Test Case	Dependencies
SRS-F-002	The system shall allow an authenticated user to log out and terminate the current session.	Medium	AC: Session is cleared and protected functions are inaccessible post-logout. Test: TC-Auth-02	—
SRS-F-003	The system shall assign users a role (Student / Faculty / Admin) and enforce role-based access.	High	AC: A Student cannot access Faculty or Admin functions; each role reaches only permitted functions. Test: TC-Auth-03	RBAC
SRS-F-004	The system shall allow an Administrator to create, update, activate, and deactivate user accounts.	High	AC: Account changes are enforced on the next login attempt. Test: TC-Auth-04	Admin only
SRS-F-005	The system shall allow users to change their own password after verifying the current password.	Medium	AC: Password is updated; the old password no longer grants access. Test: TC-Auth-05	—

4.2 Student Profile Management

Req ID	Requirement	Priority	Acceptance Criteria / Test Case	Dependencies
SRS-F-006	The system shall allow an Administrator to register a new student with name, SRN, date of birth, contact details, and programme.	High	AC: Valid student record is stored and retrievable by SRN. Test: TC-Stu-01	Authentication required
SRS-F-007	The system shall allow an Administrator to update or deactivate an existing student profile.	Medium	AC: Updated fields are persisted; deactivated students cannot log in. Test: TC-Stu-02	Ownership check
SRS-F-008	The system shall allow a student to view their own profile details.	High	AC: Student sees only their own record; another student's record is inaccessible. Test: TC-Stu-03	Access control
SRS-F-009	The system shall allow an Administrator to search and filter students by name, SRN, programme, or semester.	High	AC: Search returns only matching records. Test: TC-Stu-04	—

4.3 Course & Enrollment Management

Req ID	Requirement	Priority	Acceptance Criteria / Test Case	Dependencies
SRS-F-010	The system shall allow an Administrator to create and manage courses with a code, title, credits, and assigned faculty.	High	AC: Course is stored and appears in the course list. Test: TC-Crs-01	—
SRS-F-011	The system shall allow an Administrator to enroll students into courses for a given semester.	High	AC: Enrolled student appears in the course roster. Test: TC-Crs-02	Depends on F-006, F-010
SRS-F-012	The system shall allow an Administrator or Faculty to view the roster of enrolled students per course.	Medium	AC: Roster shows correct enrollment. Test: TC-Crs-03	—

4.4 Grade Management

Req ID	Requirement	Priority	Acceptance Criteria / Test Case	Dependencies
SRS-F-013	The system shall allow Faculty to enter or update grades (marks or letter grades) for enrolled students in their assigned courses.	High	AC: Grade is stored and visible to the student and administrator. Test: TC-Grd-01	Enrollment required

Req ID	Requirement	Priority	Acceptance Criteria / Test Case	Dependencies
SRS-F-014	The system shall automatically compute GPA for a semester and CGPA across all semesters based on grades and credit weights.	High	AC: Computed GPA/CGPA matches the expected value from the grade sheet. Test: TC-Grd-02	Depends on F-013
SRS-F-015	The system shall allow a student to view their grades and computed GPA/CGPA.	High	AC: Student sees correct, up-to-date grade data. Test: TC-Grd-03	Access control
SRS-F-016	The system shall restrict grade entry to the Faculty assigned to the course; other faculty cannot modify those grades.	High	AC: Unauthorized grade modification attempt is denied. Test: TC-Grd-04	RBAC

4.5 Attendance Management

Req ID	Requirement	Priority	Acceptance Criteria / Test Case	Dependencies
SRS-F-017	The system shall allow Faculty to record daily attendance (Present / Absent / Late) for enrolled students per course session.	High	AC: Attendance entry is stored with date and session details. Test: TC-Att-01	Enrollment required
SRS-F-018	The system shall automatically calculate each student's attendance percentage per course.	High	AC: Percentage equals (sessions present / total sessions) × 100, rounded to two decimal places. Test: TC-Att-02	Depends on F-017
SRS-F-019	The system shall flag students whose attendance falls below the configured threshold (default: 75%).	Medium	AC: Students below threshold are marked with a warning indicator. Test: TC-Att-03	Configurable threshold
SRS-F-020	The system shall allow a student to view their own attendance records and percentage per course.	Medium	AC: Student sees only their own attendance. Test: TC-Att-04	Access control

4.6 Report Generation

Req ID	Requirement	Priority	Acceptance Criteria / Test Case	Dependencies
SRS-F-021	The system shall generate a per-student academic report containing profile, semester-wise grades, GPA/CGPA, and attendance summary.	High	AC: Report data matches stored records. Test: TC-Rpt-01	Depends on F-014, F-018
SRS-F-022	The system shall allow export of student reports to a CSV or plain-text file.	Medium	AC: Exported file is readable and contains correct data. Test: TC-Rpt-02	—
SRS-F-023	The system shall allow an Administrator to generate course-level reports summarizing class performance statistics (average, highest, lowest grade; average attendance).	Medium	AC: Statistics match manual computation from stored data. Test: TC-Rpt-03	Admin only

5. Non-Functional Requirements

Req ID	Requirement	Category	Priority	Acceptance / Measurement
SRS-NF-001	Core operations (login, grade lookup, attendance entry) shall complete within 2 seconds on the target hardware.	Performance	High	Response time ≤ 2 s in all TC-Perf tests.

Req ID	Requirement	Category	Priority	Acceptance / Measurement
SRS-NF-002	Computed GPA/CGPA and attendance percentages shall be consistent with stored raw data at all times.	Data Integrity	High	Values verified against raw records in TC-INT-01 / TC-INT-02.
SRS-NF-003	The interface shall use consistent navigation so a first-time user can locate and view a student record without external help.	Usability	Medium	Usability walkthrough: TC-UX-01.
SRS-NF-004	The codebase shall be organized into modular C/C++ modules: auth, student, course, grades, attendance, reports.	Maintainability	Medium	Modules compile independently; changes in one module require no unrelated edits. Code review.
SRS-NF-005	The system shall compile and run correctly on Windows 10/11 (MSVC) and Ubuntu 20.04+ (GCC).	Portability	Medium	Build succeeds and core workflows pass on both platforms. TC-Compat-01.
SRS-NF-006	The system shall remain operational during scheduled use periods; no data shall be lost on normal shutdown.	Reliability	High	No data loss observed in TC-Rel-01 (graceful shutdown and restart test).
SRS-NF-007	The system shall handle up to 5,000 student records and 50 concurrent sessions without degradation.	Scalability	Medium	Load test with 5,000 records meets NF-001 response target. TC-Scale-01.

5.1 Security

5.1.1 Security Objectives

- Protect student academic data and personal information from unauthorized access or modification.
- Ensure only authorized roles can create, modify, or delete records.
- Maintain confidentiality and integrity of stored credentials.

5.1.2 Security Requirements

Req ID	Requirement	Priority	Acceptance Criteria / Test Case
SRS-SR-001	The system shall require valid authentication before granting access to any protected function.	High	Unauthenticated access is rejected. Test: TC-Sec-01
SRS-SR-002	The system shall enforce role-based authorization for Student, Faculty, and Administrator operations.	High	Unauthorized role attempts are denied. Test: TC-Sec-02
SRS-SR-003	The system shall store passwords using a secure one-way hash (e.g., bcrypt or SHA-256 with salt); plaintext passwords shall never be stored.	High	Inspection confirms passwords are stored only as hashes. Test: TC-Sec-03
SRS-SR-004	The system shall validate all user-provided inputs before processing or storing (type, length, allowed characters).	High	Invalid inputs are rejected with an appropriate error message. Test: TC-Sec-04
SRS-SR-005	The system shall maintain an audit log of all create, update, and delete operations on student records, capturing user, timestamp, and action.	Medium	Audit log entries are present for all record-modifying operations. Test: TC-Sec-05

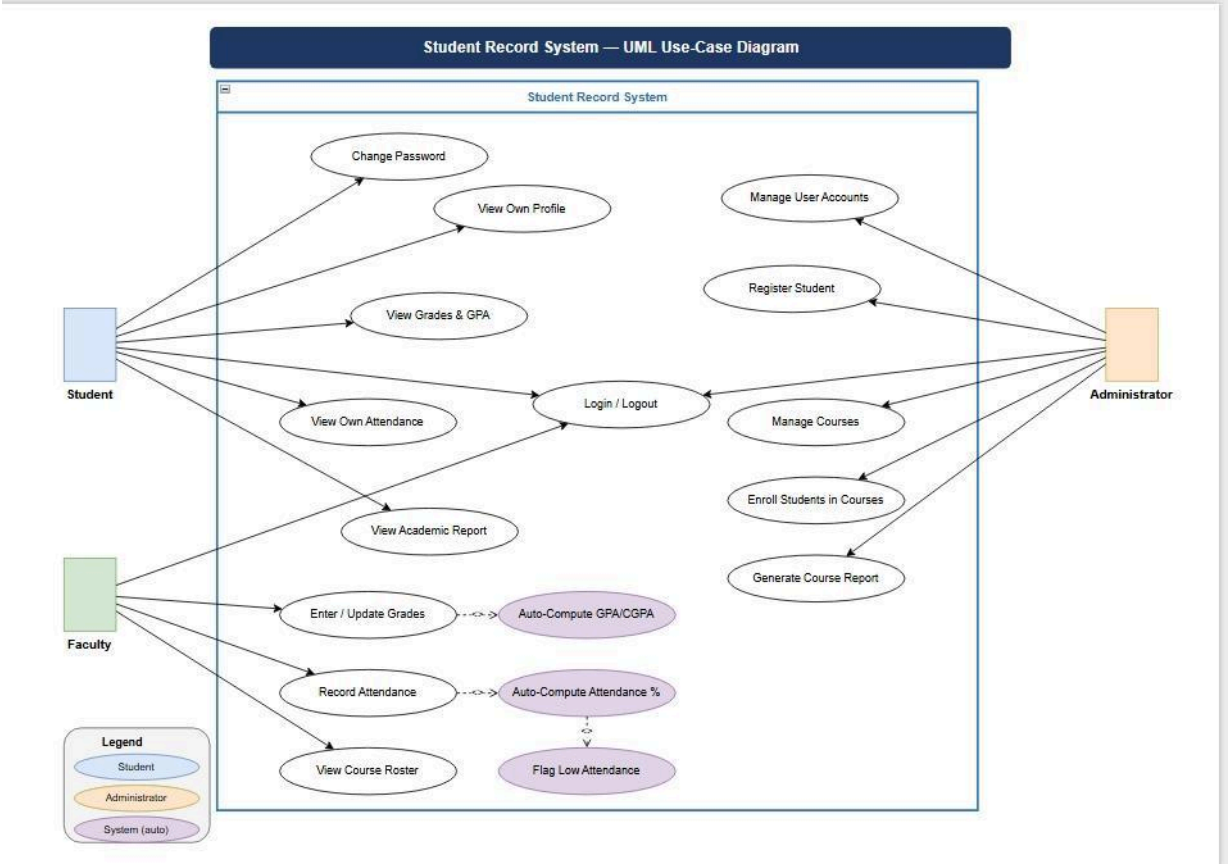
6. Quality Attributes & Acceptance Tests

The system will be considered acceptable when all high-priority functional requirements pass their tests, no critical security requirement fails, grade/GPA computations and attendance percentages remain consistent with raw data, and the stated performance, usability, and portability targets are met.

Acceptance scope:

- Authentication and role-based access control
- Student profile creation, search, and update
- Course and enrollment management
- Grade entry and GPA/CGPA computation
- Attendance recording and percentage calculation
- Report generation and export
- Administration and user management
- Security, performance, and portability

7. UML Use-Case Diagram



Actor summary:

- Student — views own profile, grades, attendance, and reports.
- Faculty — enters grades and attendance for assigned courses; views course summaries.
- Administrator — manages users, courses, enrollments, and generates institution-level reports.
- System — auto-computes GPA/CGPA and attendance percentages; generates audit logs.

8. Requirements Traceability Matrix (RTM)

Req ID	Short Description	Section	Module	Test Case	Status
SRS-F-001	Login with valid credentials	4.1	Auth & User Mgmt	TC-Auth-01	N
SRS-F-002	Logout & session termination	4.1	Auth & User Mgmt	TC-Auth-02	N

Req ID	Short Description	Section	Module	Test Case	Status
SRS-F-003	Role assignment & RBAC enforcement	4.1	Auth & User Mgmt	TC-Auth-03	N
SRS-F-004	Admin: manage user accounts	4.1	Auth & User Mgmt	TC-Auth-04	N
SRS-F-005	User: change own password	4.1	Auth & User Mgmt	TC-Auth-05	N
SRS-F-006	Admin: register new student	4.2	Student Profile	TC-Stu-01	N
SRS-F-007	Admin: update/deactivate student profile	4.2	Student Profile	TC-Stu-02	N
SRS-F-008	Student: view own profile	4.2	Student Profile	TC-Stu-03	N
SRS-F-009	Admin: search & filter students	4.2	Student Profile	TC-Stu-04	N
SRS-F-010	Admin: create & manage courses	4.3	Course & Enrollment	TC-Crs-01	N
SRS-F-011	Admin: enroll students in courses	4.3	Course & Enrollment	TC-Crs-02	N
SRS-F-012	View course roster	4.3	Course & Enrollment	TC-Crs-03	N
SRS-F-013	Faculty: enter/update grades	4.4	Grade Management	TC-Grd-01	N
SRS-F-014	Auto-compute GPA/CGPA	4.4	Grade Management	TC-Grd-02	N
SRS-F-015	Student: view grades & GPA	4.4	Grade Management	TC-Grd-03	N
SRS-F-016	Restrict grade entry to assigned faculty	4.4	Grade Management	TC-Grd-04	N
SRS-F-017	Faculty: record attendance	4.5	Attendance	TC-Att-01	N
SRS-F-018	Auto-compute attendance %	4.5	Attendance	TC-Att-02	N
SRS-F-019	Flag low-attendance students	4.5	Attendance	TC-Att-03	N
SRS-F-020	Student: view own attendance	4.5	Attendance	TC-Att-04	N
SRS-F-021	Generate per-student academic report	4.6	Reports	TC-Rpt-01	N
SRS-F-022	Export report to CSV/text	4.6	Reports	TC-Rpt-02	N
SRS-F-023	Admin: course-level performance report	4.6	Reports	TC-Rpt-03	N
SRS-NF-001	Response time ≤ 2 s	5	Performance	TC-Perf-01	N
SRS-NF-002	GPA/attendance data consistency	5	Data Integrity	TC-INT-01	N
SRS-NF-003	Consistent UI navigation	5	Usability	TC-UX-01	N
SRS-NF-004	Modular C/C++ codebase	5	Maintainability	Code Review	N
SRS-NF-005	Windows & Linux portability	5	Portability	TC-Compat-01	N
SRS-NF-006	No data loss on shutdown	5	Reliability	TC-Rel-01	N
SRS-NF-007	5,000 records / 50 sessions	5	Scalability	TC-Scale-01	N
SRS-SR-001	Require authentication	5.1.2	Security	TC-Sec-01	N
SRS-SR-002	Enforce RBAC	5.1.2	Security	TC-Sec-02	N
SRS-SR-003	Hash passwords	5.1.2	Security	TC-Sec-03	N
SRS-SR-004	Validate all inputs	5.1.2	Security	TC-Sec-04	N
SRS-SR-005	Maintain audit log	5.1.2	Security	TC-Sec-05	N